Machine Translation on Myanmar Dialects

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Outline

- Myanmar Dialects
- Statistical Machine Translation
- Neural Machine Translation
- Weighted Finite State Transducer based Machine Translation
- Transfer and Triangulation on Myanmar Dialects
- Conclusion

Myanmar Dialects

- Dialect refers to a variety of a language that is a characteristics of a particular group of the language's speakers.
- Dialects or varieties of a particular language are closely related, and despite their differences, are most often largely mutually intelligible, especially if close to one another on the dialect continuum.
- In the Republic of the Union of Myanmar, there are many ethnical groups, and dialectal varieties exist within the standard Burmese language.
- Speakers of the standard Burmese may find the dialects hard to follow
- To address this problem, we are developing Burmese-Myeik and Burmese-Rakhine and Burmese-Dawei dialectal corpora.

Myanmar Language Classification

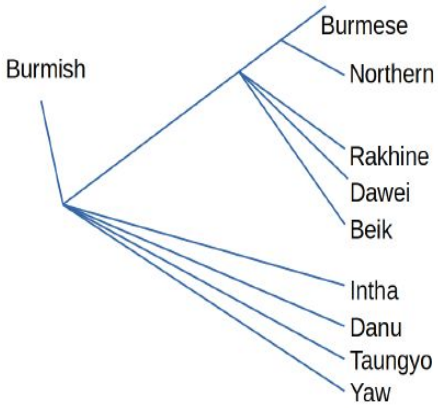


Fig. 1. Classification of Burmese (adapted from Bradley 1997: 38-39 [7]).

Introduction

- Myanmar language includes a number of mutually intelligible Myanmar dialects, with a largely uniform standard dialect used by most Myanmar standard speakers
- Speakers of the standard Myanmar may find the dialects hard to follow.
- The alternative phonology, morphology, and regional vocabulary cause some problems in communication.

Main Motivation

- Machine translation has so far neglected the importance of properly handling the spelling, lexical and grammar divergences of Dialects in Myanmar.

Objective

- to develop Burmese dialect machine translation prototype through the various experiments.
- Statistical or neural approaches are usually used in machine translation system.
- Although there are many data resources for other languages, there is no available parallel corpus for low resource languages, especially Myanmar language (Burmese) and Burmese Dialects.
- we have to build parallel corpora for Burmese dialectal languages.
- Various experiments have been tested based on statistical and neural machine translation approaches between standard Burmese and dialectal languages to achieve higher translation performance.

Contribution

- Parallel corpora building for Burmese dialects.
- The 1st SMT results of Phrase-based Machine Translation (PBSMT), Hierarchical Phrase-based Statistical Machine Translation (HPBSMT) and Operation Sequence Model (OSM) on Burmese dialects.
- The 1st NMT results of three important neural network architectures; RNN, Transformer and CNN on Burmese dialects.
- Implementation of Weighted Finite State Transducer (WFST) based machine translation system for Burmese dialects and results with various alignment techniques
- Transfer and Triangulation pivot translation results for the Beik-Burmese-Dawei, Dawei-Burmese-Beik, Beik-Burmese-Rakhine, and Rakhine-Burmese-Beik

Myanmar Dialects and Standard Myanmar Language

Rakhine

Myanmar: အဘွား နေမကောင်း ပါ
Rakhine : အဘောင်သျှင် နီမကောင်း ပါ
English : “Grandmother is not feeling well”

Dawei

Myanmar: ဒီ ကောင်မလေး လှလွန်း တယ်
Dawei : သယ်ဝယ်သား လှ ဗြင်း တယ်
English : “This girl is very beautiful”

Myeik

Myanmar: ကျွန်တော် မနက်ဖြန် နိုင်ငံခြားသွားမယ်
Beik : ငါ မောလင်း နိုင်ငံခြား သွားမယ်
English : “I will go foreign tomorrow”

Segmentation Rules for Myanmar Dialects

- root word and suffixe(s) are separated such as “စား ဗျာယ်”, “စား ပီးဗျာယ်”, “စား ဖို့ဗျာယ်”. Here, “စား” (“eat” in English) is a root word and the others are suffixes for past and future tenses.
- plural nouns are identified by following particle. We also put a space between noun and the following particle, for example a Rakhine word “ကလိန်ုမေ့ချ့ တိ” (ladies) is segmented as two words “ကလိန်ုမေ့ချ့” and the particle “တိ”. In
- particles describe the type of noun, and used after number or text number. For example, a Rakhine word “အကြ့ိုစ့ေ့ နှစ်ခတ်” (“two coins” in English) is segmented as “အကြ့ိုစ့ေ့ နှစ် ခတ်”.
- compound nouns are considered as one word and thus, a Rakhine compound word “ဖ့ေသာ + အိတ်” (“money” + “bag” in English) is written as one word “ဖ့ေသာအိတ်” (“wallet” in English).
- adverb words such as “အဂယောင့်” (“really” in English), “အမြန်” (“quickly” in English) are also considered as one word.

Cyclone Mocha may have killed ‘hundreds’ in Myanmar’s Rakhine

Communications problems are making it difficult to assess the extent of the damage caused by the massive storm.

အိမ်ရှင်မတို့ချင်းအိမ်မှုကိစ္စမဟုသုတ ဖလှယ်ကြစို့

Pont Pont · 20m ·

စစ်တွေ အခြေနေထင်တာပဲ
ဖတ်ရတာတော့ နားမလည်ဘူး 👇

David Htun is with **Chun Twan Maung** and **6 others**.
46m ·

မိဇီ ပန်းခြံနောက်က 09896157403 09669882927 ကျော်သီဟထွန်း (or) gp chay အေဖုန်းကိုဆက်သွယ်ပြီး တခေါက်ကယ်ပီးကပ်ပါ အိမ်တ အိမ်လုံး မြုပ်ဖို့ကန်နီလို့ သူတယောက်တည်း ဖုန့်ဆက်ပြီး ကူညီတောင်းနီလို့ပါ ဟူး

ဖုန်းဆက်သွယ်လို့ တဖုန်းလေ့ဆက်မရဖစ်နိပါရယ်...ဟူး

Dialect Corpus

- ASEAN-MT Parallel Corpus (without name Entity Tag)
- 18K sentences Myanmar-Rakhine Parallel Corpus
- 9K sentences Myanmar-Dawei sentences Parallel Corpus
- 10K sentences Myanmar-Beik sentences Parallel Corpus

Statistical Machine Translation

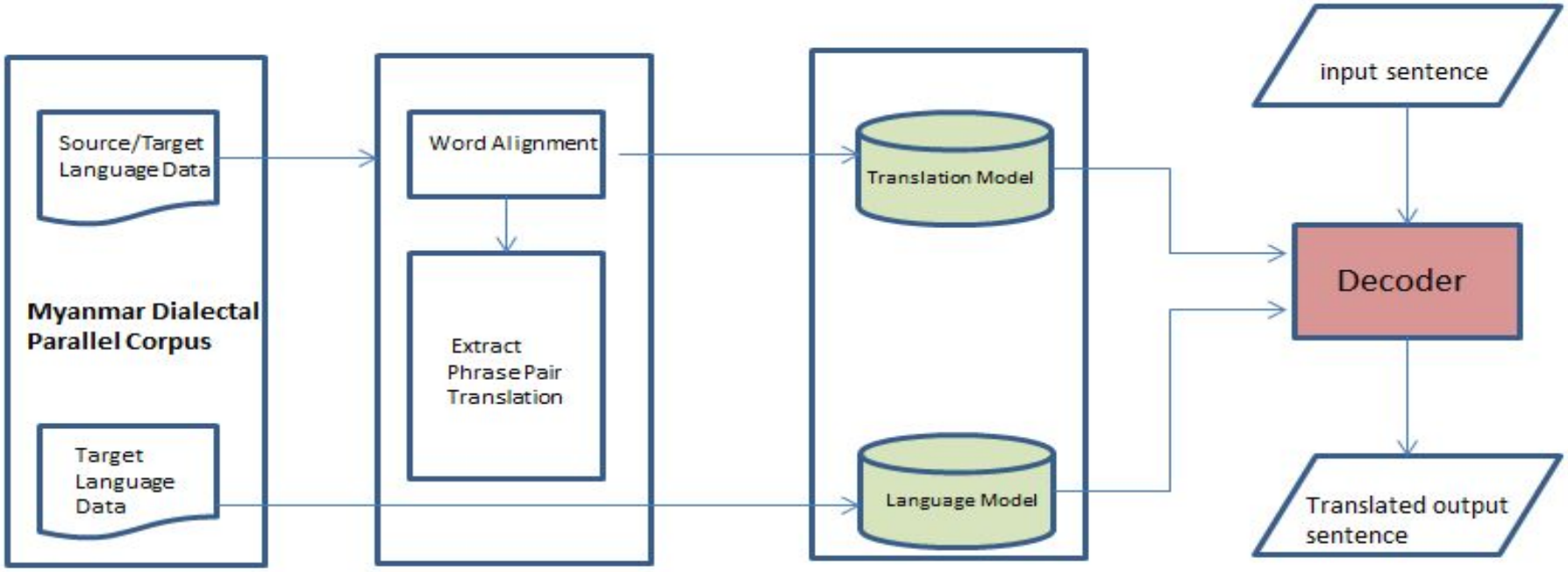


Fig2 Overview architecture of Statistical Machine Translation

Statistical Machine Translation

Phrase Based SMT

- based on phrasal units .
- A phrase is simply a contiguous sequence of words and generally, not a linguistically motivated phrase

Hierarchical phrase-based SMT

- based on synchronous context-free grammar.
- to be learned from a corpus of unannotated parallel text.
- re-ordering is represented explicitly rather than encoded into a lexicalized re-ordering model
- makes the approach particularly applicable to language pairs that require long-distance re-ordering during the translation process

Operation sequence model

- combines the benefits of two state-of-the-art SMT frameworks named n-gram-based SMT and phrase-based SMT.
- generates source and target units and does not have spurious ambiguity that is based on minimal translation units
- motivates better reordering mechanism that uniformly handles local and non-local reordering and strong coupling of lexical generation and reordering.
- It means that OSM can handle both short and long distance reordering.

SMT Results

	PBSMT	HPBSMT	OSM
Myanmar-Rakhine	83.34	83.37	**83.83**
Rakhine-Myanmar	84.27	84.06	**85.18**
Myanmar-Beik	54.60	54.40	**55.11**
Beik-Myanmar	70.01	69.89	**70.54**
Myanmar-Dawei	44.80	45.44	**45.58**
Dawei-Myanmar	60.78	60.47	**63.22**

Work 1 (SMT)

Thazin Myint Oo, Ye Kyaw Thu, Khin Mar Soe, Thepchai Supnithi, "Statistical Machine Translation of Myanmar Dialects", Journal of Intelligent Informatics and Smart Technology, April 1st Issue, 2020, pp. 14-26. (Submitted December 21, 2019; accepted March 6, 2020; revised March 18, 2020; published online April 30, 2020) ("**the Workshop Paper Award**" at the First Joint Myanmar-Thai NLP/AI R&D Workshop, 2019, Chiang Mai, Thailand)

- The main problem for myanmar machine translation is lack of data resources. we also developed three Myanmar Dialect corpora based on the Myanmar language of ASEAN MT corpus. They are Myanmar-Rakhine (18K), Myanmar-Myeik (10K) and Myanmar-Dawei (9K) parallel corpora.
- The 10 folds cross-validation experiments were carried out using phrase-based, hierarchical phrase-based, and the operation sequence model with two types of segmentation; word and syllable units were studied.
- The results show that all three statistical machine translation approaches give higher and comparable BLEU and RIBES scores between Myanmar and three dialects (Rakhine, Dawei and Myeik) in both directions.
- The OSM approach achieved the highest BLEU and RIBES scores among three approaches for both word and syllable segmentations. Moreover, we found that syllable segmentation is appropriate for translation quality comparing with word level segmentation results.

Neural Machine Translation

Recurrent Neural Network

- Encoder layer consists of a bi-directional RNN followed by a stack of uni-directional RNNs. first layer produces a forward sequence of hidden states, some non-linear function, such as a Gated Recurrent Unit (GRU) or Long Short Term Memory (LSTM) cell. The reverse RNN processes the source sentence from right to left: and the hidden states from both directions are concatenated. The hidden state can incorporate information from both tokens to the left, as well as tokens to the right.
- Decoder consists of an RNN to predict one target word at a time through a state vector

Transformer

- Encoder has three stages. First is input words are projected into an embedded vector space. Second stage is multi-head self-attention and individual attention blocks compute the scaled dot-product attention with different linear projection. Third stage is position-wise fully connected feed-forward network is used.
- Decoder generate one word at a time from left to right. It has five stages. The first two stages are similar to the encoder. Third has multi-head self-attention, which unlike in the encoder, forces to attend only to past words but also the final representation generated by the encoder. Fourth stage is another position-wise feed forward network. Finally, a soft-max layer allows to map target word scores into target word probabilities.

Convolutional Neural Network

- convolutional operations and also dispensed with recurrence. Input embedding are again augmented with explicit positional encodings. The convolutional encoder applies a set of (stacked) convolutions To increase the context window captured by the encoder architecture, multiple layers of convolutions are stacked. To maintain sequence length across multiple stacked convolutions, inputs are padded with zero vectors.
- Decoder is similar to the encoder but adds and attention mechanism to every layer. The output of the target side convolution

Work 2 (NMT)

Thazin Myint Oo, Ye Kyaw Thu, Khin Mar Soe and Thepchai Supnithi, **"Neural Machine Translation between Myanmar (Burmese) and Dawei (Tavoyan)",** In Proceedings of the 18th International Conference on Computer Applications (ICCA 2020), Feb 27-28, 2020, Yangon, Myanmar, pp. 219-227.

- This work explores the first evaluation of the quality of neural machine translation between Myanmar (Burmese) and Dawei (Tavoyan). We also developed Myanmar-Dawei parallel corpus (around 9K sentences) based on the Myanmar language of ASEAN MT corpus.
- Recurrent Neural Network (RNN) and Transformer with syllable segmentation. We also investigated various hyper-parameters such as batch size, learning rate and cell types (GRU and LSTM).
- LSTM cell type with RNN architecture is the best for Dawei-Myanmar and Myanmar-Dawei neural machine translation. Myanmar to Dawei NMT achieved comparable results with PBSMT and HPBSMT.
- Dawei to Myanmar RNN machine translation performance achieved higher BLEU scores than PBSMT (+1.06 BLEU) and HPBSMT (+1.37 BLEU) even with the limited parallel corpus.

Work 3 (NMT)

Thazin Myint Oo, Ye Kyaw Thu, and Khin Mar Soe, **"Neural Machine Translation between Myanmar (Burmese) and Rakhine (Arakanese)",** In Proceedings of the Sixth Workshop on NLP for Similar Languages, Varieties and Dialects, NAACL-2019, June 7th 2019, Minneapolis, United States, pp. 80-88.

- recurrent neural networks (RNN), transformer, and convolutional neural networks (CNN). The systems were evaluated on a Myanmar-Rakhine parallel text corpus developed by us.
- two types of word segmentation schemes for word embeddings were studied: Word-BPE and Syllable-BPE segmentation. Our experimental results clearly show that the highest quality NMT and statistical machine translation (SMT) performances are obtained with Syllable-BPE segmentation for both types of translations.
- If we focus on NMT, we find that the Transformer with Word-BPEsegmentation outperforms CNN and RNN for both Myanmar-Rakhine and Rakhine-Myanmar translation. However, CNN with Syllable-BPE segmentation obtains a higher score than the RNN and transformer.

Comparison SMT and NMT for Rakhine Myanmar Language pair

Segmentation	OSM		RNN		Transformer		CNN	
	My-Rk	Rk-My	My-Rk	Rk-My	My-Rk	Rk-My	My-Rk	Rk-My
Syllable-BPE	82.71	**84.36**	82.03	83.98	82.85	82.65	**83.75**	84.08
Word-BPE	77.12	75.27	60.31	46.47	73.39	72.95	69.03	72.24

- Transformer with Word-BPEsegmentation outperforms CNN and RNN for both Myanmar-Rakhine and Rakhine-Myanmar translation. However, CNN with Syllable-BPE segmentation obtains a higher score than the RNN and transformer.
- Compare with SMT Myanmar-Rakhine CNN model gets greater score OSM.

Comparison SMT and NMT for Dawei Myanmar Language pair

	PBSMT	HPBSMT	RNN	Transformer
Myanmar-Dawei	44.8	45.44	44.68	43.03
Dawei-Myanmar	60.78	60.47	**61.84**	60.44

- LSTM cell type with RNN architecture is the best for Dawei-Myanmar and Myanmar -Dawei neural machine translation. Myanmar to Dawei NMT achieved comparable results with PBSMT and HPBSMT.
- Dawei to Myanmar RNN machine translation performance achieved higher BLEU scores than PBSMT (+1.06 BLEU) and HPBSMT (+1.37 BLEU) even with the limited parallel corpus.

Work 4 (WFST)

Thazin Myint Oo, Thitipong Tanprasert, Ye Kyaw Thu and Thepchai Supnithi, "Syllable-to-Syllable and Word-to-Word Transducers for Burmese Dialect Translation", In Proceedings of the 17th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP 2022), Nov 5 to 7, 2022, Chiang Mai, Thailand, pp. 1-6

- Weighted Finite State Transducers (WFST) can be very efficient to implement Burmese dialects translation.
- We illustrate this on two Burmese dialect language pairs, Burmese-Beik and Burmese-Rakhine.
- In this study, we examine syllable and word segmentation schemes and their effect on alignment and transducing between dialect language pairs.
- We performed alignments with Anymalign, fastalign, pialign, Hieralign, eflomal and GIZA++ approaches and implemented WFST based machine translation system with OpenFst library.
- From the overall results, syllable segmentation achieved higher BLEU and chrF scores for Burmese-Rakhine and Rakhine-Burmese translations.
- However, word segmentation achieved better translation performance for Burmese-Beik and Beik-Burmese translation directions. Alignment techniques fastalign, Hieralign, eflomal and GIZA++ are working well for low-resource Burmese dialects.

https://ieeexplore.ieee.org/document/9960259

Work 5 (Pivot Based SMT)

Thazin Myint Oo, Thitipong Tanprasert, Ye Kyaw Thu, Thepchai Supnithi, "Transfer and Triangulation Pivot Translation Approaches for Burmese Dialects," in *IEEE Access*, vol. 11, pp. 6150-6168, 2023, doi: 10.1109/ACCESS.2023.3236804. (Received 15 October 2022, accepted 27 December 2022, date of publication 13 January 2023, date of current version 20 January 2023.)

- Parallel corpora for the languages of Myanmar (Beik, Burmese, Rakhine) are extremely scarce but a necessary requirement for machine translation R&D.
- Previous studies have proved that pivoting leads to better translation quality if the bridge language is closely related to the source and target language pair.
- The baseline study is conducted based on the three major approaches of machine translation; Weighted Finite State Transducer (WFST), Phrase-Based Statistical Machine Translation (PBSMT) and Deep Recurrent Neural Network (Deep-RNN)
- Based on the baseline results, this work mainly investigated the pivot language technique for PBSMT with Burmese dialects.
- two different pivot translation methods: transfer (sentence level) and triangulation (phrase level).
- present the experimental results on Dawei-Beik, Beik-Dawei translations and Beik-Rakhine, Rakhine-Beik translation via Burmese.
- Both the transfer and triangulation approaches outperformed the baseline (direct translation), specifically for the Rakhine-Beik language pair.

Baseline Experiment

- Weighted Finite State Transducer Based Machine Translation (WFSTs)
- Phrase Based Statistical Machine Translation (PBSMT)
- Deep Recurrent Neural Network (Deep-RNN)

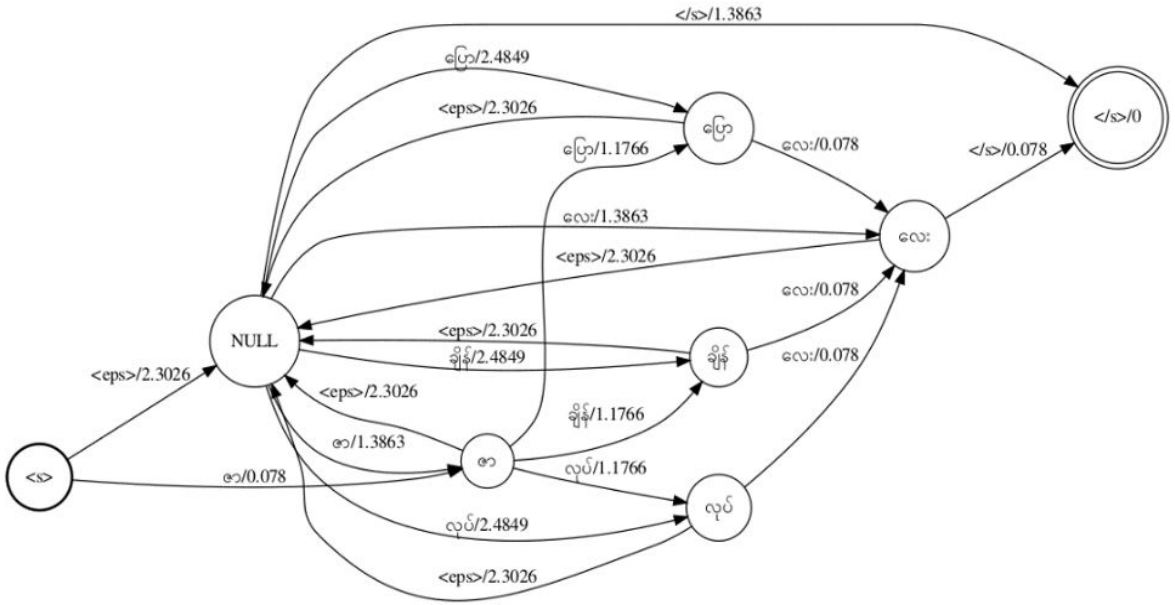
Weighted Finite State Transducer Based MT

Weighted Finite State Transducer (WFST) algorithms have been effectively applied to a wide range of tasks in natural language processing, including parsing and language machine translation, speech recognition, speech synthesis, optical character recognition, spelling checking and image processing

I. Weighted Finite State Automata (WFSA) Based Language Model
II. WFST Based Translation Model
III. Composing of Language Model and Translation Model WFSTs

Weighted Finite State Automata (WFSA) Based Language Model

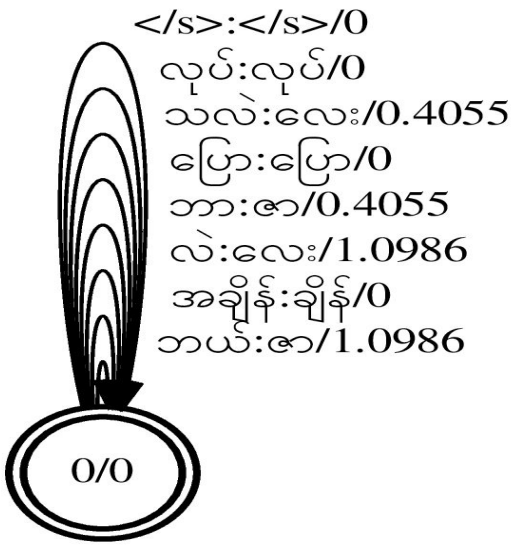
- Weighted Finite State Automaton (WFSA) are able to work like a language model (i.e. the smoothed *n-gram* languages model) by expressing sets of strings with corresponding score over them.
- Edge labels are "word/score", where the score is represented as a negative log probability. States are labeled with the context of Rakhine words r_{t-1} that the represent. The "NULL"state represents unigrams. The "<eps>" represents an $\in$ edge, it allows fall back from the unigram state to the bigram state.



WFST Based Translation Model

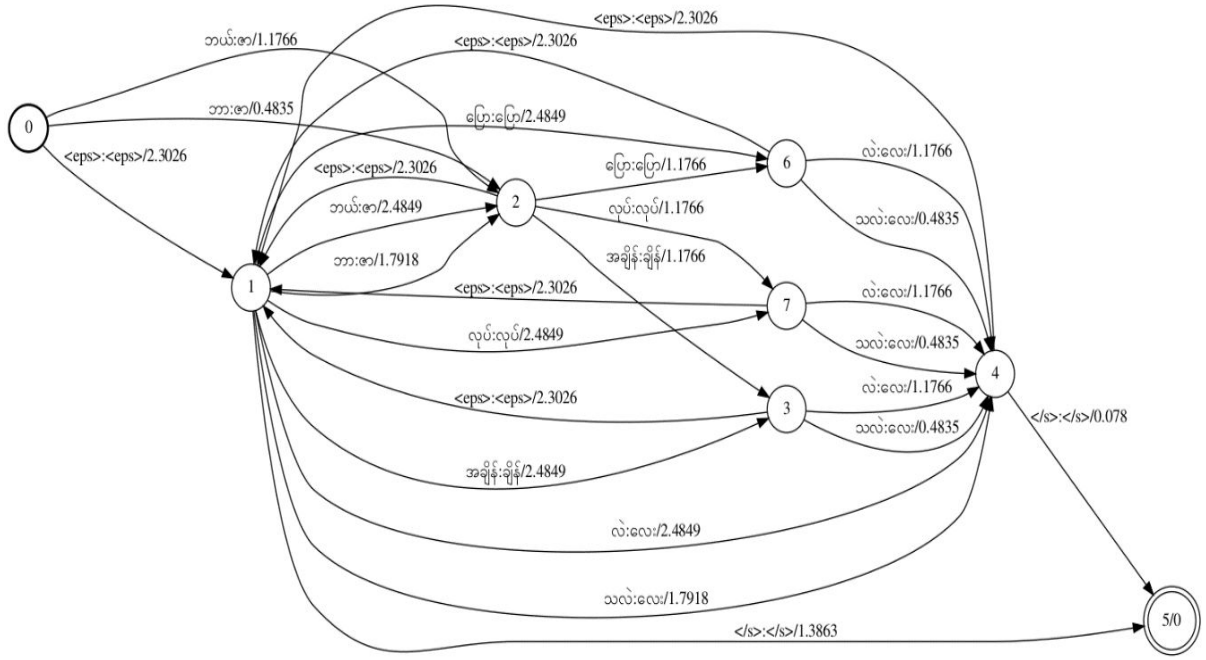
- WFSTs are basically similar to WFSAs but use two labels on every edge.
- specifies a mapping between two sets of strings. To give a very simple translation model even simpler than IBM Model 1: it calculates $P(R|B)$ for translation from Burmese (B) to Rakhine (R) by taking one b_t at a time and independently calculates the translation probability of the corresponding word r_t

$$P(R|B) = \prod_{t=1}^{|B|} P(r_t|b_t)$$



Composing of Language Model and Translation Model WFSTs

- To find the best translation, we can combine the translation model probability $P(R|B)$ and language model probability $P(R)$ scores.
- apply the "composition" operation of the two transducers to create a new WFST model. This composition can be express as $T_{P(R|B)} \circ T_{P(R)}$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Implementation of WFST Based MT

- WFST-based MT system is implemented using and OpenFST and Anymalign
- Anymalign allows multiple languages to be aligned simultaneously from sentence-aligned parallel corpora.
- It used sub-sentential alignment method for extracting high quality multi-word alignments from the corpora https://anymalign.limsi.fr/
- The calculations for unigram and bigram probabilities based on the aligned phrases was coded with Python and the wrapper programs for WFST based translation with OpenFST commands were coded with bash script.
- The WFST composition process of language model and translation model tranducers were done by applying *fstcompose* operation of the OpenFST framework.
- The best translation result was selected by *shortest-path* algorithm based on edit-distance between strings (i.e. edit-distance semi-ring).

Phrase-Based Statistical Machine Translation

- is based on phrasal unit
- simple phrase-based translation model that consists of phrase-pair probabilities extracted from a corpus, a basic reordering model, and an algorithm to extract the phrases to build a phrase-table.
- find the best translation $\hat{e}$ that maximizes the translation probability $\boldsymbol{P}(f)$ given the source sentences

$$\hat{e} = argmax_e \boldsymbol{P}(e|f)$$

- Applying the Bayes' rule, we can factorize into three parts

$$\boldsymbol{P}(e|f) = \frac{\boldsymbol{P}(e)}{\boldsymbol{P}(f)} \boldsymbol{P}(f|e)$$

- The final mathematical formulation of the phrase-based model is given according to (5):

$$argmax_e \boldsymbol{P}(e|f) = argmax_e \boldsymbol{P}(f|e)\boldsymbol{P}(e)$$

Moses SMT

- Moses toolkit (version 2.1.1) open source toolkit for statistical machine translation to train the direct source-target and transfer SMT.
- The word-segmented source language was aligned with the word-segmented target language using GIZA++ alignment was symmetrized using the grow-diag-final heuristic.
- lexicalized reordering model was trained using the msd-bidirectional-fe option.
- KenLM language model with modified Kneser-Ney discounting.

Deep Recurrent Neural Network (Deep-RNN)

- RNN-based encoder-decoder model with attention,
- Training procedure of RNN contained two steps, encoding and decoding (i.e. two RNNs).
- Given the source sentence $w_s \in \mathbb{B}_+^{n_s}$, where each element represents the index of the source word, the encoder first takes some piece of data as the input and transforms it into a fixed-length vector (i.e. a sequence of embeddings), denoted by $\mathbf{X}_s^1 \in \mathbb{R}^{\mathrm{n_s} \times d}$, where n_s is the length of the source sentence, and d is the dimension of the model. Then, the encoder calculates a sequence of hidden vectors by $\mathbf{H}_s^1 \in \mathbb{R}^{\mathrm{n_s} \times d}$ as follows:
$$\mathbf{H}_s^1[\mathrm{i}] = \mathrm{RNN}\left(\mathbf{H}_s^1[\mathrm{i}-1], \mathbf{X}_s^1[\mathrm{i}]\right).$$
- The function RNN can be either the gated recurrent unit (GRU) or the long short-term memory (LSTM). The input to the next layer is
$$\mathbf{X}_s^2 = \mathbf{H}_s^1.$$
- Repeating this procedure, we obtain the final representation of the source sentence $\mathbf{Y}_s = \mathbf{X}_s^{L_s+1}$ where L_s is the number of layers of the encoder.

Deep Recurrent Neural Network (Deep-RNN)

- Decoder generates the target sentence $w_t \in \mathbb{B}_+^{n_t}$ as follows:

$$\mathbf{w}_t[\mathrm{j}] = argmax\left(softmax\left(FFNN(\mathbf{Y}_t[\mathrm{j}])\right)\right),$$

- n_t is the length of the target sentence, FFNN is a feed-forward neural network, $\mathbf{Y}_t = \mathbf{X}_t^{L_t+1}$ is the representation of the target sentence, and L_t is the number of decoder layers.
- For the *l*-th layer, the output $\mathbf{X}_t^{l+1}$ is calculated as follows:

$$\mathbf{X}_t^{l+1} = f(\mathbf{C}^l, \mathbf{H}_t^l),$$ where is f a non-linear function.

- The calculation for $\mathbf{H}_t^l$ is as follows:

$$\mathbf{H}_t^1[\mathrm{j}] = \mathrm{RNN}(\mathbf{H}_t^1[\mathrm{j}-1], \mathbf{X}_t^l[\mathrm{j}]).$$

- The vector $\mathbf{C}^l[\mathrm{j}]$ is the weighted average of the columns of $\boldsymbol{Y_s}$:

$$\boldsymbol{C}^l[j] = \sum_{i=1}^{n_s} softmax\left(\mathbf{H}_t^l[\mathrm{j}]\right)^\top \mathbf{Y}_s \mathbf{Y}_s[i].$$

Marian NMT framework

- used the model types `--type s2s` for RNN with attention model, three encoding and decoding layers, LSTM for both encoding and decoding cells

```
--type s2s
--max-length 200
--workspace 5500
--enc-depth 3 --enc-type alternating --enc-cell lstm --enc-cell-depth 4
--dec-depth 3 --dec-cell lstm --dec-cell-base-depth 4
--dec-cell-high-depth 2
--tied-embeddings --layer-normalization --skip
--mini-batch-fit
--valid-mini-batch 32
--valid-metrics cross-entropy perplexity bleu
--valid-freq 5000 --save-freq 5000 --disp-freq 500
--dropout-rnn 0.3 --dropout-src 0.3 --exponential-smoothing
--early-stopping 10
```

Training process overview of RNN-attention based NMT

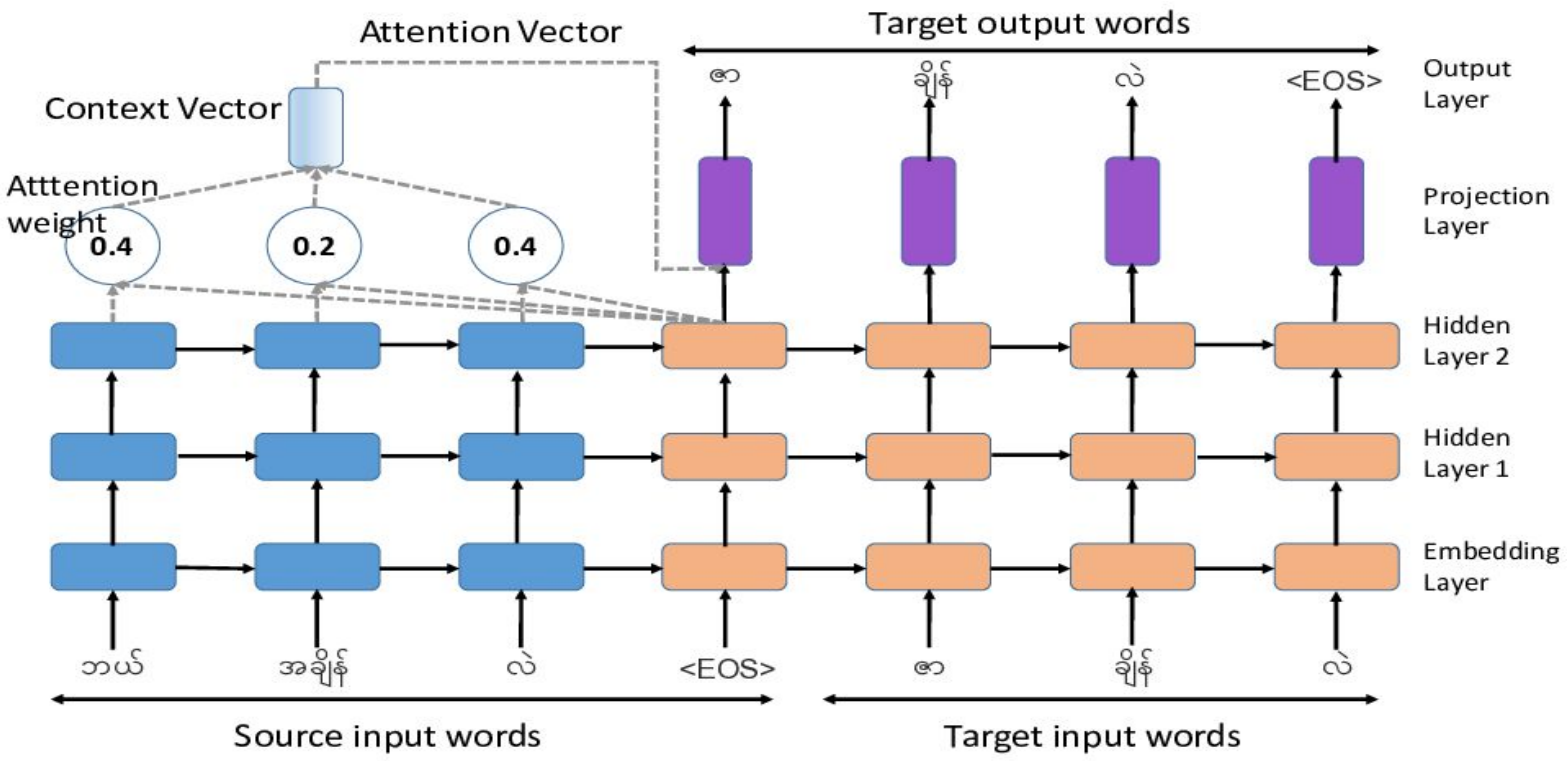


Fig3 Overview Training Process of RNN-attention

Evaluation

- The de facto standard automatic evaluation metric bilingual evaluation understudy (**BLEU**) score measures the precision of *n-gram* (overall $n \leq 4$ in our case) with respect to a reference translation with a penalty for short translations.
- Intuitively, the BLEU score measures the adequacy of the translation and large BLEU scores are better.

$$BLEU_n = BP \prod_{i=1}^{n} precision_i$$

- ***chrF*** is a character *n-gram* F-score for the automatic evaluation of the MT

$$chrF\beta = (1 + \beta^2)\frac{chrP.chrR}{\beta^2.chrP + chrR}$$

where $chrP$ and $chrR$ represent the character *n-gram* precision and recall, arithmetically averaged over all *n-grams*

Baseline Result

TABLE I BLEU SCORES OF WFST, PBSMT, AND DEEP-RNN METHODS FOR BEIK-DAWEI AND BEIK-RAKHINE LANGUAGE PAIRS (THE SCORES HIGHER THAN DIRECT TRANSLATION ARE HIGHLIGHTED IN BOLD)

Source-Target	WFST	PBSMT	Deep-RNN
Beik-to-Dawei	9.77	**18.02**	13.60
Dawei-to-Beik	10.53	**20.37**	12.67
Beik-to-Rakhine	20.06	**36.70**	19.48
Rakhine-to-Beik	16.31	**30.20**	22.06

TABLE II *CHRF* SCORES OF WFST, PBSMT, AND DEEP-RNN METHODS FOR BEIK-DAWEI AND BEIK-RAKHINE LANGUAGE PAIRS (THE SCORES HIGHER THAN DIRECT TRANSLATION ARE HIGHLIGHTED IN BOLD).

Source-Target	WFST	PBSMT	Deep-RNN
Beik-to-Dawei	32.23	**53.45**	31.84
Dawei-to-Beik	35.69	**50.92**	31.44
Beik-to-Rakhine	52.81	**75.89**	45.06
Rakhine-to-Beik	46.35	**64.22**	38.33

Pivot Based SMT

A common solution to data sparsity in the field is to pivot the translation through a third language (called pivot or bridge language) for which there exists abundant parallel corpora with the source and target languages .Two pivot strategies have been proposed

- Transfer or Sentence Level Pivoting
- Triangulation or Phrase Level Pivoting

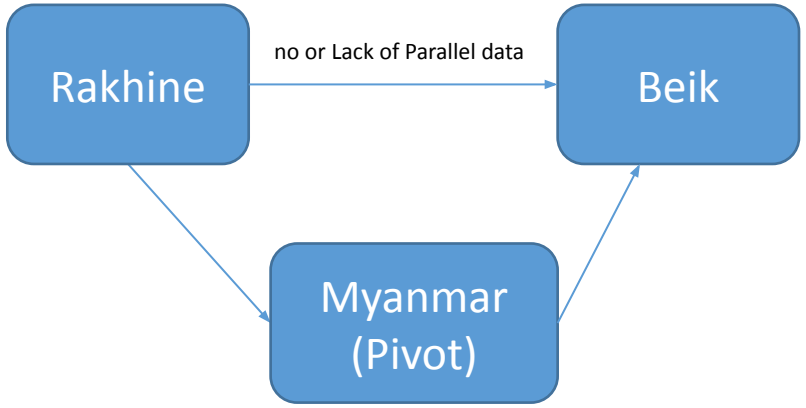


Fig4 Rakhine to Beik Pivot-Based Translation

Transfer or Sentence Level Pivoting

Transfer method (sentence-level pivoting or cascade), in which the pivot language is used as an interface between two separate phrase-based MT systems in which we first translate the source sentence to the pivot language, and then translate the pivot language sentence to the target language

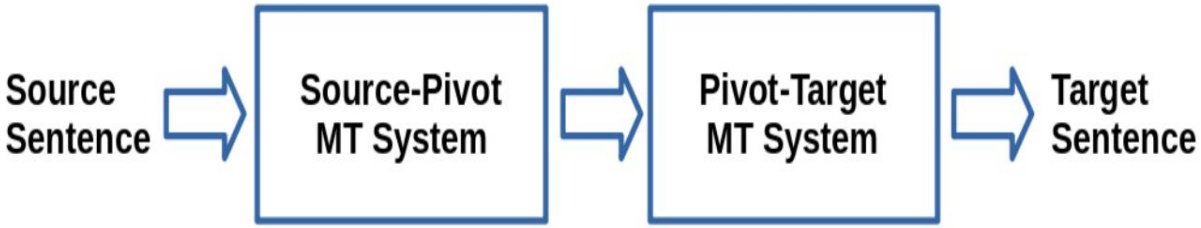


Fig5 Transfer method or Sentence-level pivoting

Transfer or Sentence Level Pivoting

- translate source sentence s into n pivot sentences $p_1, p_2, \cdots, p_n$ using a source-pivot PBSMT system.
- Each $p_i (i = 1, \cdots, n)$ has eight scores: language model probability of the target language, two-phase translation probabilities, two lexical translation probabilities, word penalty, phrase penalty, and linear reordering penalty.

$$h^p_{i_1}, h^p_{i_2}, h^p_{i_3}, \cdots, h^p_{i_8}.$$

- Second, we translated each p_i into n target sentences $t_{i_1}, t_{i_2}, \cdots, t_{i_n}$ using a pivot-target PBSMT system.
- Each $t_{ij} (j = 1, \cdots, n)$ also has eight scores, denoted as $h^t_{ij_1}, h^t_{ij2}, h^t_{ij3}, \cdots, h^t_{ij8}$.

$$s \rightarrow p_i\left(h^p_{i1}, h^p_{i2}, h^p_{i3}, \cdots, h^p_{i8}\right) \rightarrow t_{ij}\left(h^t_{ij1}, h^t_{ij2}, h^t_{ij3}, \cdots, h^t_{ij8}\right)$$

We define the score of $t_{ij}, S(t_{ij})$, as

$$S(t_{ij}) = \sum_{m=1}^{8} (\lambda^p_m h^p_{im} + \lambda^t_m h^t_{im})$$

where λ^p_m and λ^t_m are weights set by performing minimum error rate training

- Finally, the selection of the highest-scoring target sentence $\hat{t}$ is $\hat{t} = argmax_{t_{ij}} S(t_{ij})$.

Triangulation or Phrase Level Pivoting

It is based on two pre-trained models (source-pivot and pivot-target translations) and it induces a new source-target translation model

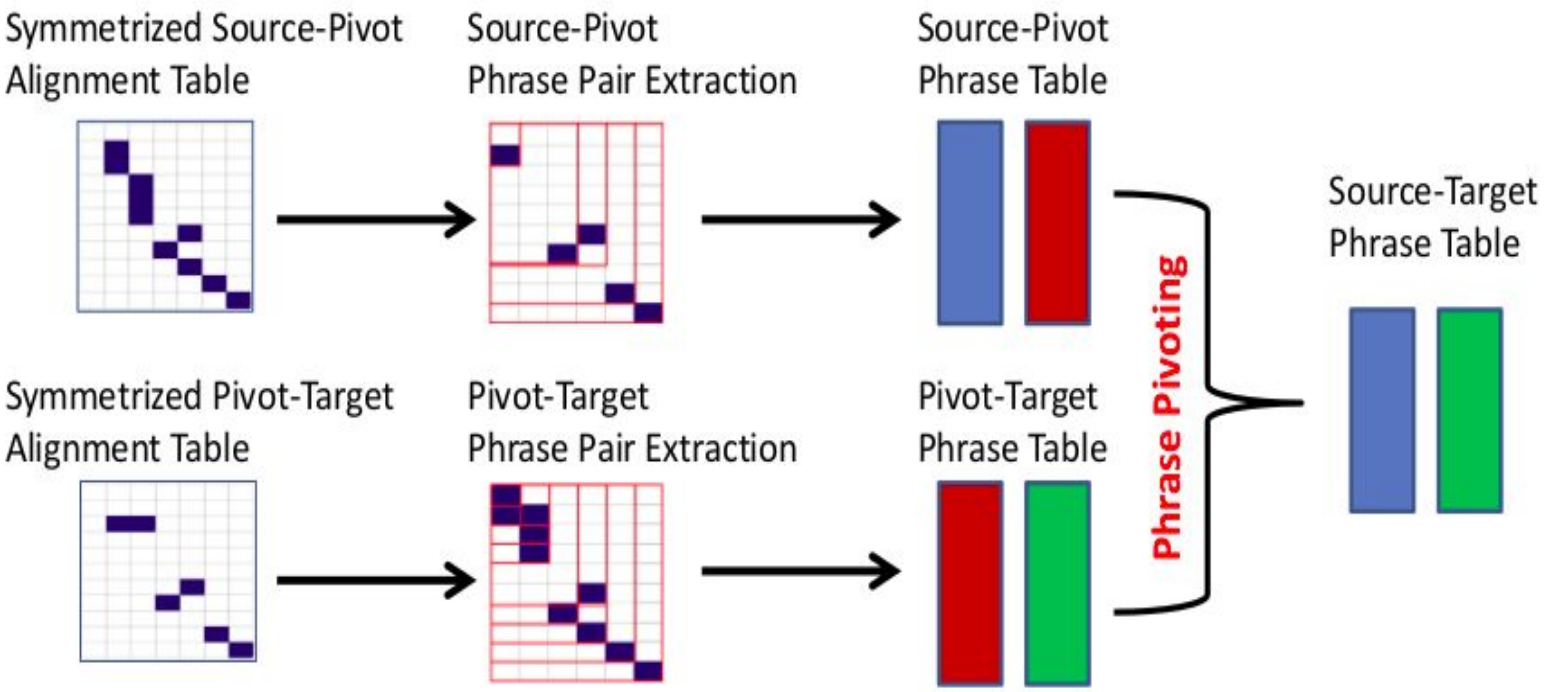


Fig6 Triangulation or Phrase level Pivoting

- Building statistical machine translation systems for both source-pivot and pivot-target
- A source-target phrase table T_{fe} is constructed by combining a source-pivot T_{fp} and pivot target phrase tables T_{pe}
- Combining phrase alignment: for each phrase pair (f, e), we can compute an alignment $\hat{a}$ as the most frequent alignment obtained by combining source-pivot a_{fp} and pivot-target alignments a_{pe} across all pivot phrases p as follows:

$$\hat{a} = (f, e) | \exists P : (f, p) \in a_{fp} \wedge (p, e) \in a_{pe}$$

- Estimation of Lexical weighting probability: First, word translation scores $\hat{w}_{fe}$ are approximated by marginalizing over the pivot words:

$$\hat{w}_{fe}(e|f) = \sum_{p} w_{fp}(p|f) \cdot w_{pe}(e|p)$$

- Next, given a (triangulated) phrase pair (f , e) with alignment $\hat{a}$, let $\hat{a}_f$ = e | (f , e) ∈ $\hat{a}$; the lexical weighting probability

$$\hat{lex}_{fe}(e|f, \hat{a}) = \prod_{f \in f} \frac{1}{|\hat{a}_{f,:}|} \sum_{e \in \hat{a}_{f,:}} \hat{w}_{fe}(e|f)$$

- Scoring: direct and reverse phrase and lexical translation probabilities

Corpus Statistics (Beik-Dawei)

Table 1. Average number of sentences, average number of words, and percentage of the OOV in the training, development, and test splits for each language used in our pivot experiments for Beik-Burmese-Dawei and Dawei-Burmese-Beik. The average sentence length for each corpus has also been described.

Language	Data-set	Sentence (Avg.)	Word Unit (Avg.)	% of OOV
Beik (Avg. length: 6.34)	Training	5459.80	34549.60	-
	Development	500.00	3213.00	-
	Test	662.20	4195.3	9.27
Burmese (Pivot language, Avg. length: 6.39)	Training	5459.80	34858.70	-
	Development	500.00	3253.60	-
	Test	662.20	4234.70	8.85
Dawei (Avg. length: 5.99)	Training	5459.80	32669.30	-
	Development	500.00	3023.40	-
	Test	662.20	3965.90	14.43

Corpus Statistics (Beik-Rakhine)

Table 2. Average number of sentences, average number of words and percentage of the OOV in the training, development, and test splits for each language used in our pivot experiments for Beik-Burmese-Rakhine and Rakhine-Burmese-Beik. The average sentence length for each corpus has also been described.

Language	Data-set	Sentence (Avg.)	Word Unit (Avg.)	% of OOV
Beik (Avg. length: 6.57)	Training	8850.00	58110.9	-
	Development	800.00	5292.7	-
	Test	1072.00	7043.4	8.23
Burmese (Pivot language, Avg. length: 6.52)	Training	8850.00	57743.00	-
	Development	800.00	5214.3	-
	Test	1072.00	6993.7	8.36
Rakhine (Avg. length: 6.43)	Training	8850.00	56917.7	-
	Development	800.00	5170.3	-
	Test	1072.00	6897.00	9.18

Results on BLEU

Table 3. Average BLEU scores of direct translation, transfer method, and triangulation method for Beik-Dawei and Beik-Rakhine language Pairs (the scores higher than direct translation are highlighted in bold).

src-trg	Source → Target	Source → Pivot → Target	
	Direct Translation	*Transfer*	*Triangulation*
Beik → Dawei	18.02	17.98	**18.23**
Dawei → Beik	20.37	19.67	20.27
Beik → Rakhine	36.70	36.64	**37.62**
Rakhine → Beik	30.20	**31.29**	**30.46**

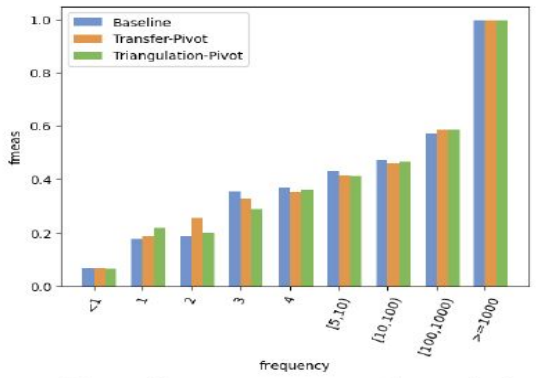
Results on chrF

Table 4. chrF scores of direct translation, transfer method, and triangulation method for Beik-Dawei and Beik-Rakhine language pairs (the scores higher than direct translation are highlighted in bold).

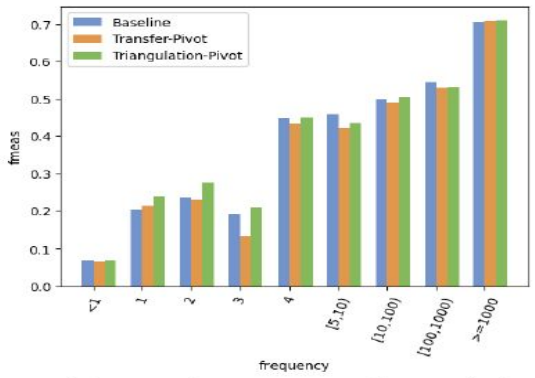
src-trg	Source → Target	Source → Pivot → Target	
	Direct Translation	*Transfer*	*Triangulation*
Beik → Dawei	53.45	**53.63**	**53.61**
Dawei → Beik	50.92	50.71	**51.24**
Beik → Rakhine	75.89	**76.27**	74.75
Rakhine → Beik	64.22	**65.49**	**64.28**

Bucketed Word Analysis

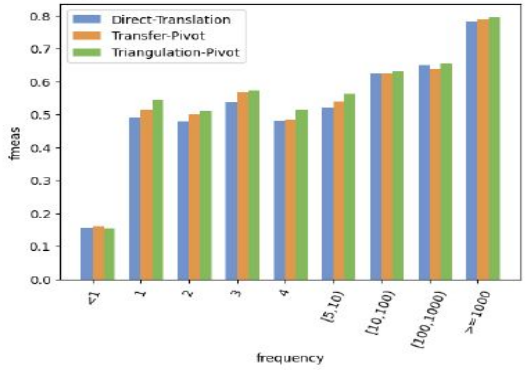
- Comprehensive using the compare-mt tool
- Test set unigram F-1 scores of occurrence in the predicted sentences based on their frequencies in the training corpus



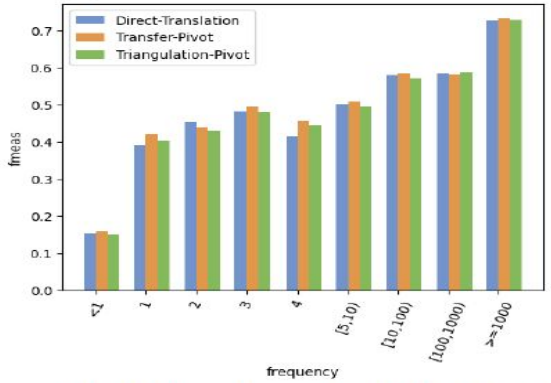
(a) Beik-Burmese-Dawei translation



(b) Dawei-Burmese-Beik translation



(c) Beik-Burmese-Rakhine translation

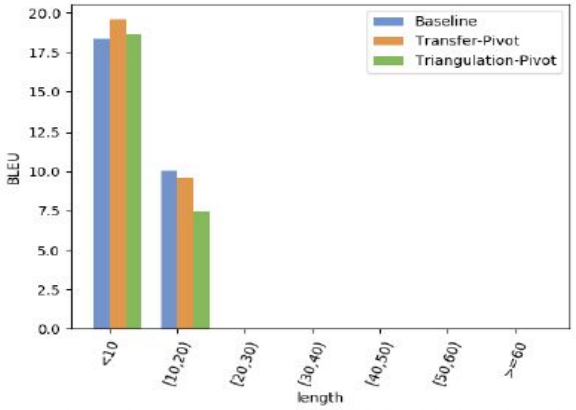


(d) Rakhine-Burmese-Beik translation

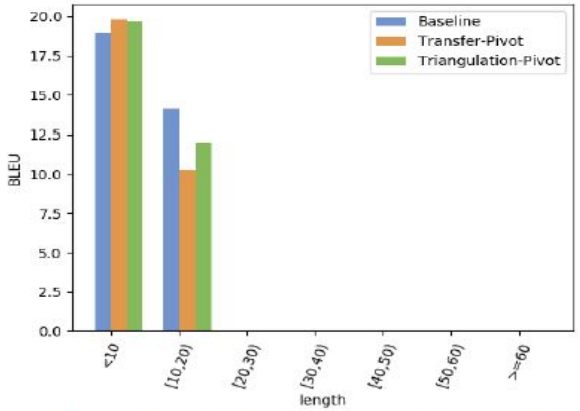
- For frequency 4, [5,10), and [10,100) ranges of *Beik-Burmese-Rakhine* and *Rakhine-Burmese-Beik* translation, both the transfer and triangulation approaches perform slightly better, and all three systems give comparable results for high-frequency words (≥ 1000).

Bucketed Sentence Analysis

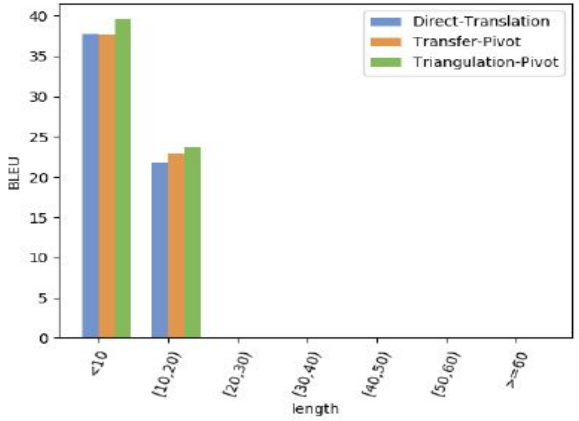
BLEU scores bucketed by the sentence length



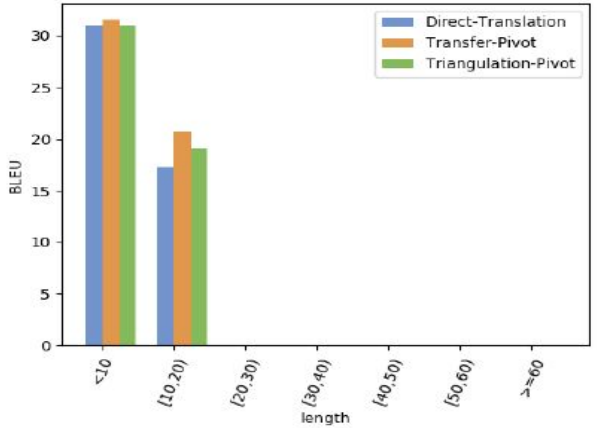
(a) Beik-Burmese-Dawei translation



(b) Dawei-Burmese-Beik translation



(c) Beik-Burmese-Rakhine translation

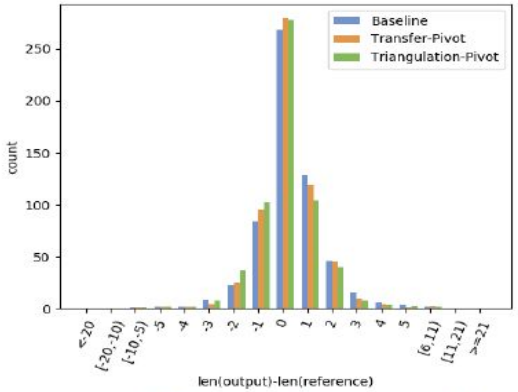


(d) Rakhine-Burmese-Beik translation

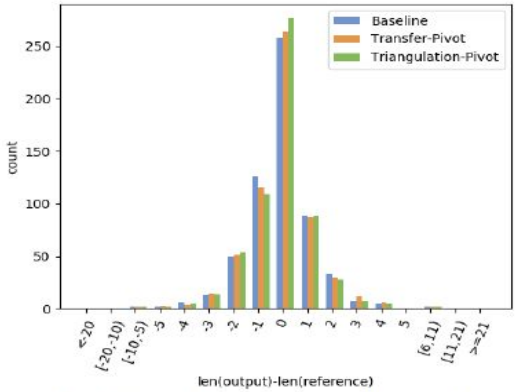
- shows that transfer and triangulation perform better for [10, 20) range long sentences for Beik-Burmese-Rakhine and Rakhine-Burmese-Beik translation directions.
- Conversely, the baseline performs better for [10, 20) range long sentences than the two pivot approaches for the Beik-Burmese-Dawei and Dawei-Burmese-Beik translations.
- Moreover, the pivot approaches were comparable or better for short (< 10) sentences than the baseline for all four translation directions.

Bucketed Sentence Analysis

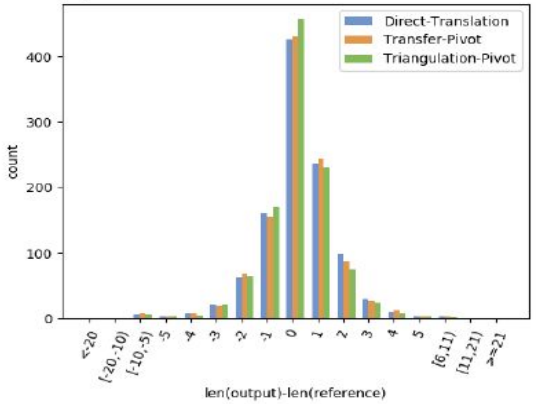
Counts of sentences by length difference between the reference and the output.



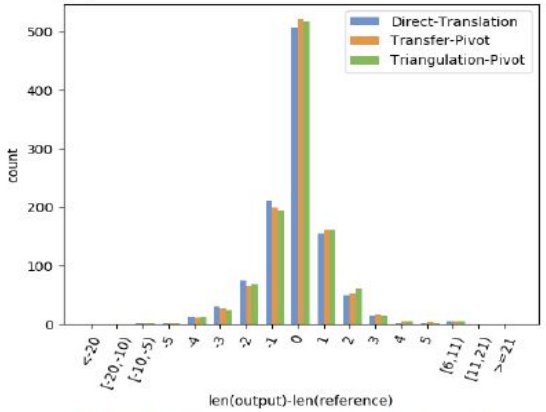
(a) Beik-Burmese-Dawei translation



(b) Dawei-Burmese-Beik translation



(c) Beik-Burmese-Rakhine translation



(d) Rakhine-Burmese-Beik translation

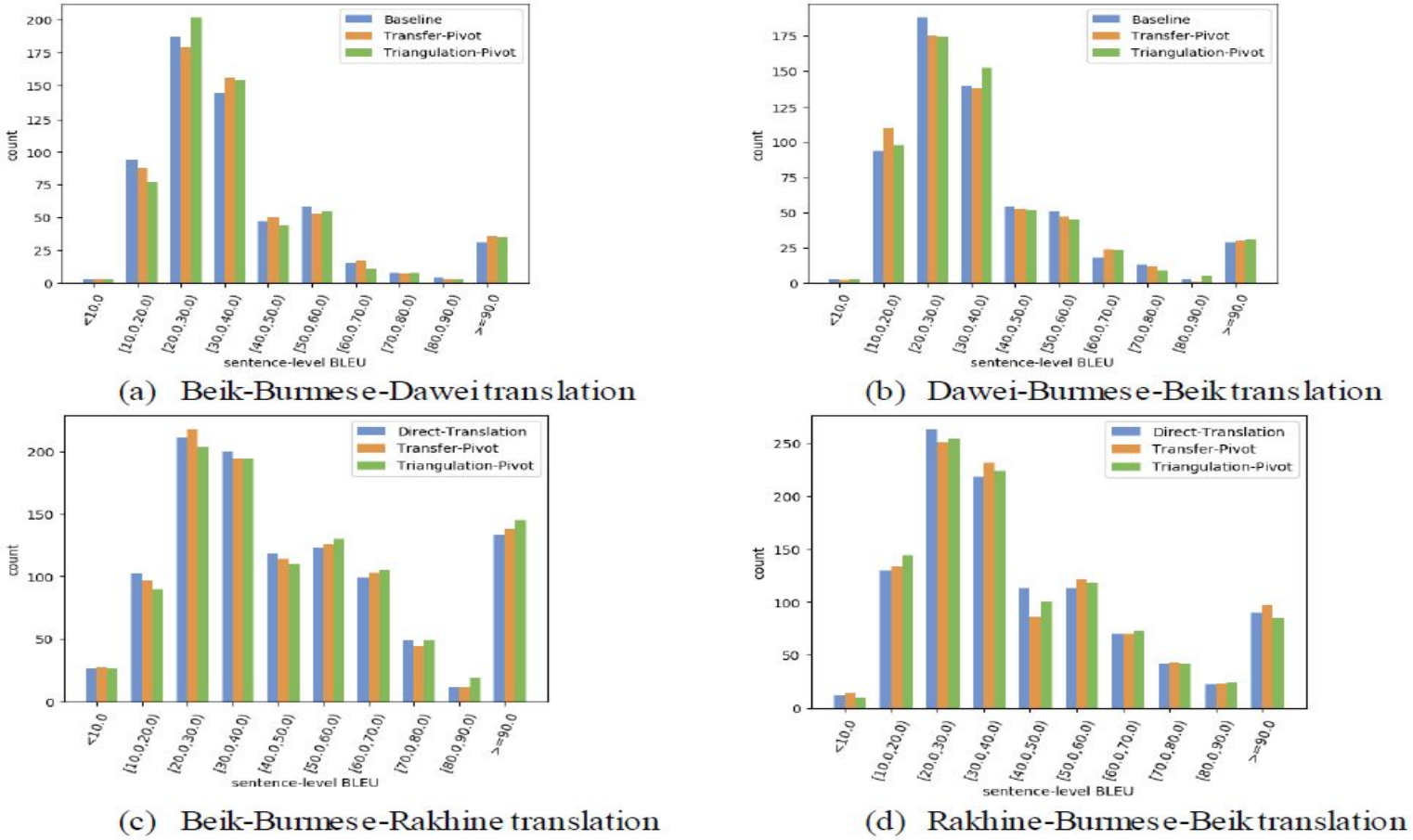
a histogram of the number of sentences that deviate from the reference output by a specific length.

A distribution that peaks around 0 suggests that the system generally matches the output length, but a flatter distribution shows that the system has problems generating sentences of the right length.

shows that both transfer and triangulation are better at generating sentences that are close to the references.

Bucketed Sentence Analysis

Counts of sentences by sentence-level BLEU bucket



(a) Beik-Burmese-Dawei translation

(b) Dawei-Burmese-Beik translation

(c) Beik-Burmese-Rakhine translation

(d) Rakhine-Burmese-Beik translation

This generates a histogram of the number of sentences with a particular score (e.g. sentence-level BLEU score).

It indicates the number of sentences of a particular accuracy each system generates.

shows that both transfer and triangulation have slightly more sentences with high BLEU scores in the range [60.0, 70.0) for all translation directions.

Length Ratio

The length ratio is the length of the output to the input, and generally, the length ratios of the transfer and triangulation approaches are approximately 1.0, which are comparable to or slightly better than the baseline

TABLE VII LENGTH RATIO (HERE, "REF" FOR REFERENCE AND "HYP" FOR HYPOTHESIS OR OUTPUT).

Source-Target	Baseline	Transfer	Triangulation
Beik-to-Dawei	1.041	1.029	1.008
	(ref=3550, hyp=3697)	(ref=3550, hyp=3651)	(ref=3550, hyp=3579)
Dawei-to-Beik	0.972	0.977	0.972
	(ref=3747, hyp=3641)	(ref=3747, hyp=3659)	(ref=3747, hyp=3641)
Beik-to-Rakhine	1.022	1.018	1.008
	(ref=6898, hyp=7046)	(ref=6898, hyp=7020)	(ref=6898, hyp=6953)
Rakhine-to-Beik	0.978	0.990	0.989
	(ref=7081, hyp=6927)	(ref=7081, hyp=7009)	(ref=7081, hyp=7002)

Translation Error Analysis

We used SCLITE version2.10 (SCLITE is part of the SCTK toolkit. Available at https://github.com/usnistgov/SCTK accessed on January 27, 2022)

Table 6. The average WER (lower is better) of direct translation, transfer, and triangulation methods for Beik-Dawei and Beik-Rakhine language pairs (the scores lower than the baseline are highlighted in bold).

src-trg	Source → Target	Source → Pivot → Target	
	Direct Translation	*Transfer*	*Triangulation*
Beik → Dawei	59.5%	**59.4%**	**58.4%**
Dawei → Beik	56.5%	56.7%	**56.0%**
Beik → Rakhine	44.9%	**44.7%**	**43.6%**
Rakhine → Beik	47.6%	**47.0%**	**47.3%**

Top Ten Confusion Pair

TABLE IX TOP 10 CONFUSION PAIRS OF TRANSFER FOR DAWEI-BURMESE-BEIK TRANSLATION (LEFT) AND TRIANGULATION FOR BEIK-BURMESE-RAKHINE TRANSLATION (RIGHT).

Transfer (Dawei-Burmese-Beik)		Triangulation (Beik-Burmese-Rakhine)	
Frequency	Reference => Hypothesis	Frequency	Reference => Hypothesis
136	မင်း ==> နင်	98	ပါလား။ ==> ။
127	ဝို ==> ကို	86	ငါ ==> ကျွန်တော်
87	ကျွန်တော် ==> ငါ	86	ရေ။ ==> ။
67	ကို ==> ဝို	80	ပါ။ ==> ။
66	ရယ် ==> ရိ	74	မင်းကို ==> ကို
55	ဒယ်ကောင်မငယ် ==> သူ	73	ကျွန်တော် ==> ငါ
49	ခင်ဗျား ==> နင်	67	ထိုမချေ ==> သူ
49	မ ==> ကို	63	လေး။ ==> ။
44	လဲ ==> ရိ	60	ပါရေ။ ==> ။
39	ဝ ==> ရ	56	ရို့ ==> သူရို့

Translation Error Analysis (word segmentation)

TABLE X EXAMPLE COMPARING SENTENCES (HYPOTHESES OF BEIK-MYANMAR-DAWEI TRANSLATION) WHERE TRIANGULATION-PIVOT PERFORMED BETTER THAN TRANSFER-PIVOT BASED ON WORD SEGMENTATION.

	Output	sentBLEU
Reference	သူ့ိုနိ့ နန်ဟ့ို မေး ကေ့ဟ့ုလား ။	-
Transfer-Pivot	သူ့ိုနိ့ **နန်ဟ့ို** မေး ဟ့ုလား ။	26.2691
Triangulation-Pivot	သူ့ိုနိ့ **နန်ဟ့ို** မေး ဟ့ုလား ။	53.7285

Translation Error Analysis (reordering)

TABLE XI EXAMPLE COMPARING SENTENCES (HYPOTHESES OF BEIK-MYANMAR-DAWEI TRANSLATION) WHERE TRIANGULATION-PIVOT PERFORMED BETTER THAN TRANSFER-PIVOT BASED ON REORDERING.

	Output	sentBLEU
Reference	နန် အဲဟ္ဝိ မဖြစ်မနေ သွား ဟ္ဝိုလား ။	-
Transfer-Pivot	နန် အဲဟ္ဝိ **သွား မဖြစ်မနေ** ဖ့ိုလား ။	34.5721
Triangulation-Pivot	နန် အဲဟ္ဝိ **မဖြစ်မနေ သွား** ဖ့ိုလား ။	64.3459

Translation Error Analysis (spelling inconsistence)

TABLE XII EXAMPLE COMPARING SENTENCES (HYPOTHESES OF BEIK-MYANMAR-DAWEI TRANSLATION) WHERE TRANSFER-PIVOT PERFORMED BETTER THAN TRIANGULATION-PIVOT BASED ON SPELLING INCONSISTENCE.

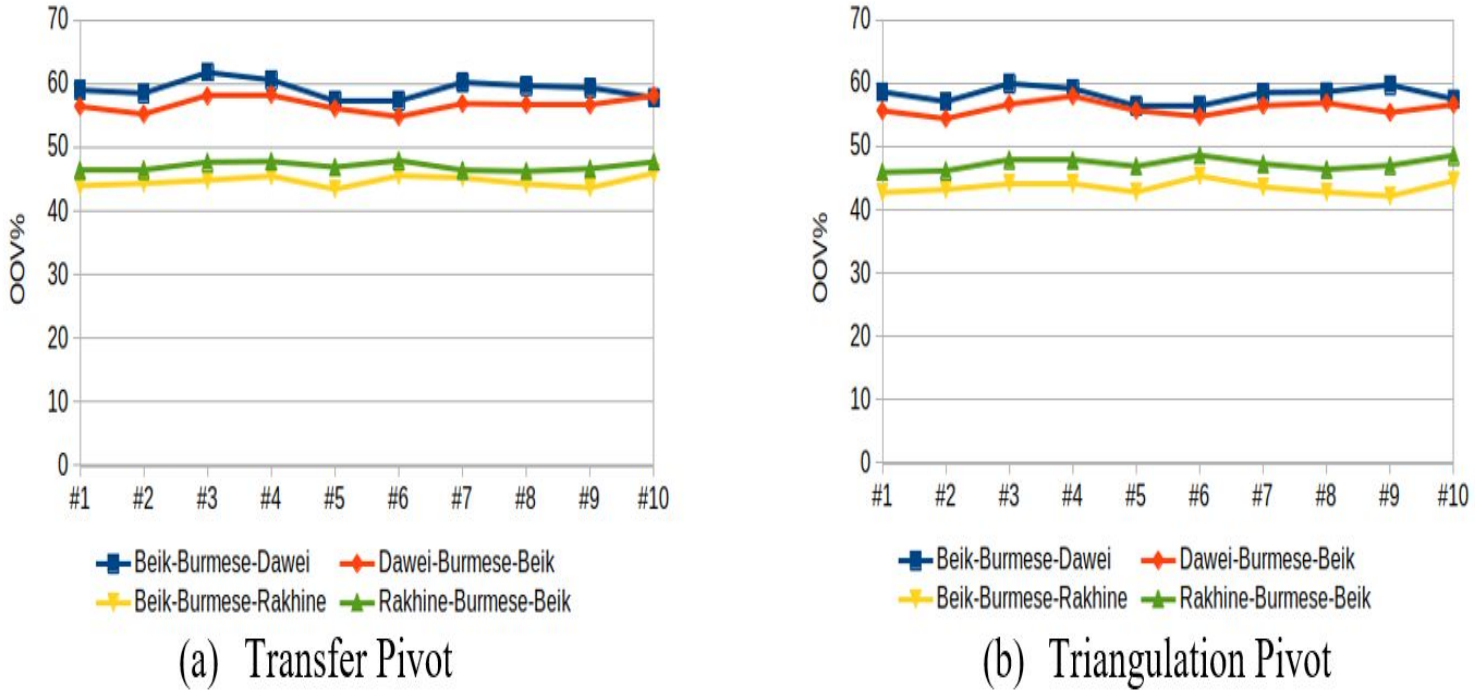
	Output	sentBLEU
Reference	မေးခွန်း မေး ဟ၂လား။	-
Transfer-Pivot	မေးခွန်း **ၜမး** ဟ၂လား။	100.0000
Triangulation-Pivot	မေးခွန်း **ၜမ** ဟ၂လား။	50.8133

Translation Error Analysis (paraphrase)

TABLE XIII EXAMPLE COMPARING SENTENCES (HYPOTHESES OF BEIK-MYANMAR-RAKHINE TRANSLATION) WHERE TRANSFER-PIVOT PERFORMED BETTER THAN TRIANGULATION-PIVOT BASED ON PARAPHRASE.

	Output	sentBLEU
Reference	ထိုမချေ မ ဉ့် ပါ ။	-
Transfer-Pivot	ထိုမချေ **မ ဉ့် ပါ** ။	100.0000
Triangulation-Pivot	ထိုမချေ **မ ဉ့်** ။	55.0695

Out of Vocabulary Percentage



indicate that the OOV% of the Beik-Dawei parallel corpus is higher than that of the Beik-Rakhine corpus, which affects the translation performance.

Increasing the size of the current parallel corpora improves the performance of the pivot systems.

Fig7 Out of Vocabulary percentage Dialect language Pair

Conclusion

SMT

- three main SMT approaches: PBSMT, HPBSMT and OSM for Burmese dialect translations.
- Good translation performance of three SMT models even with the limited dialect data.
- OSM is the best model for translation between standard Burmese and Burmese dialects.
- syllable segmentation provide higher BLEU and RIBES scores than word segmentation.

NMT

- Applied three prominent NMT systems: RNN, transformer, and CNN.
- sub-word segmentation schemes: Word-based-BPE and Syllable-based-BPE were applied.
- transformer architecture with Word-based-BPE segmentation outperforms CNN and RNN for both Burmese-Rakhine and Rakhine-Burmese translations. However, on the other hand, CNN with syllable-based-BPE obtains a higher BLEU score than RNN and transformer architectures.
- LSTM cell type with RNN architecture is the best for Dawei-Myanmar bi-directional machine translation.
- compared results between SMT, Dawei to Myanmar RNN machine translation performance achieved higher BLEU scores than PBSMT (+1.06 BLEU) and HPBSMT (+1.37 BLEU) even with the limited parallel corpus.

Conclusion

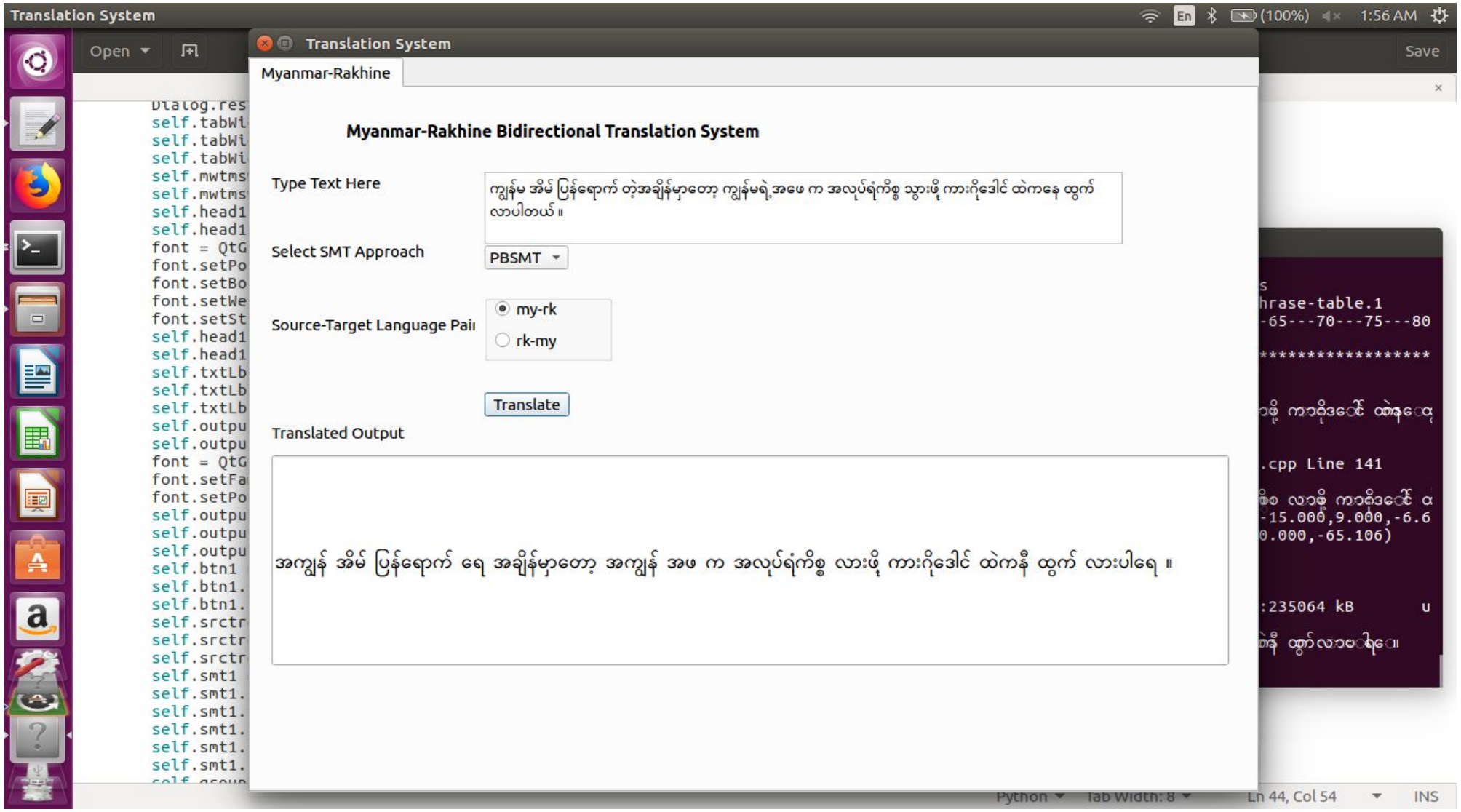
WFST

- WFST based machine translation system by combining various alignment methods and OpenFST framework.
- syllable and word segmentations and found that the results were varied based on the dialect language pairs.
- syllable segmentation achieved the best translation performance for Burmese-Rakhine and Rakhine-Burmese, word segmentation achieved the best score for Burmese-Beik and Beik-Burmese language pairs.
- results also indicate that firstalign with word segmentation and hieralign, eflomal and GIZA++ with syllable segmentation provide the best dialect translation performance.

Pivot

- the first investigation results of pivot-based SMT on low-resource Burmese dialectal languages; Dawei, Beik, and Rakhine, using Burmese as a pivot.
- The experimental results proved that both transfer and triangulation methods are applicable for the translation of Burmese dialect language pairs.

Demo


Translation System
Translation System
Myanmar-Rakhine
Myanmar-Rakhine Bidirectional Translation System
Type Text Here
ကျွန်မ အိမ် ပြန်ရောက် တဲ့အချိန်မှာတော့ ကျွန်မရဲ့အဖေ က အလုပ်ရုံကိစ္စ သွားဖို့ ကားဂိုဒေါင် ထဲကနေ ထွက် လာပါတယ် ။
Select SMT Approach
PBSMT
Source-Target Language Pai
my-rk
rk-my
Translate
Translated Output
အကျွန် အိမ် ပြန်ရောက် ရေ အချိန်မှာတော့ အကျွန် အဖ က အလုပ်ရုံကိစ္စ လားဖို့ ကားဂိုဒေါင် ထဲကနီ ထွက် လားပါရေ ။

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- ✔ **Thazin Myint Oo**, Ye Kyaw Thu**,** Khin Mar Soe, "Statistical Machine Translation between Myanmar (Burmese) and Rakhine (Arakanese)", In Proceedings of ICCA2018, February 22-23, 2018, Yangon, Myanmar, pp. 304-311.
- ✔ **Thazin Myint Oo**, Ye Kyaw Thu and Khin Mar Soe, "Neural Machine Translation between Myanmar (Burmese) and Rakhine (Arakanese)", In Proceedings of the Sixth Workshop on NLP for Similar Languages, Varieties and Dialects, NAACL-2019, June 7th 2019, Minneapolis, United States, pp. 80-88
- ✔ **Thazin Myint Oo**, Ye Kyaw Thu, Khin Mar Soe and Thepchai Supnithi, "Statistical Machine Translation between Myanmar (Burmese) and Dawei (Tavoyan)", The First Workshop on NLP Solutions for Under Resourced Languages (NSURL 2019), 11-13 September 2019, Trento, Italy
- ✔ **Thazin Myint Oo**, Ye Kyaw Thu, Khin Mar Soe and Thepchai Supnithi, "Statistical Machine Translation between Myanmar and Myeik", In Proceedings of the 12th International Conference on Future Computer and Communication (ICFCC 2020), Feb 26-28, 2020, Yangon, Myanmar, pp. 36-45
- ✔ **Thazin Myint Oo**, Ye Kyaw Thu, Khin Mar Soe and Thepchai Supnithi, "Neural Machine Translation between Myanmar (Burmese) and Dawei (Tavoyan)", In Proceedings of the 18th International Conference on Computer Applications (ICCA 2020), Feb 27-28, 2020, Yangon, Myanmar, pp. 219-227.
- ✔ **Thazin Myint Oo**, Ye Kyaw Thu, Khin Mar Soe, Thepchai Supnithi, "Statistical Machine Translation of Myanmar Dialects", Journal of Intelligent Informatics and Smart Technology, April 1st Issue, 2020, pp. 14-26. (Submitted December 21, 2019; accepted March 6, 2020; revised March 18, 2020; published online April 30, 2020) ("**the Workshop Paper Award**" at the First Joint Myanmar-Thai NLP/AI R&D Workshop, 2019, Chiang Mai, Thailand)
- ✔ **Thazin Myint Oo**, Thitipong Tanprasert, Ye Kyaw Thu, Thepchai Supnithi, "Transfer and Triangulation Pivot Translation Approaches for Burmese Dialects," in IEEE Access, vol. 11, pp. 6150-6168, 2023, doi: 10.1109/ACCESS.2023.3236804. (Received 15 October 2022, accepted 27 December 2022, date of publication 13 January 2023, date of current version 20 January 2023.)

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- ✔ Ye Kyaw Thu, **Thazin Myint Oo**, Hlaing Myat Nwe, Khaing Zar Mon, Nang Aeindray Kyaw, Naing Linn Phyo, Nann Hwan Khun and Hnin Aye Thant, "Hybrid Statistical Machine Translation for English-Myanmar: UTYCC-Team1 Submission to WAT-2021", In Proceedings of the 8th Workshop on Asian Translation (WAT2021), Aug 6, 2021, Bangkok, Thailand, pp. 83-89
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- ✔ Zar Zar Hlaing, Ye Kyaw Thu, **Thazin Myint Oo**, Mya Ei San, Sasiporn Usanavasin, Ponrudee Netisopakul and Thepchai Supnithi, "NECTEC's Participation in WAT-2021", In Proceedings of the 8th Workshop on Asian Translation (WAT2021), Aug 6, 2021, Bangkok, Thailand, pp. 74−82.
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- Dr. Khin Aye Than (Prorector, University of Computer Studies Dawei)

Myeik Myanmar Parallel Corpus

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- Dr. Thandar Win (Rector, University of Computer Studies Myeik)

THANK YOU!